WHAT IF?! EP 06

FREE QUIZ + FACT SHEET
you commented TIGER

Is Your Cat Secretly a Tiger?



Eight questions, one real fact each. Tick every one that sounds like your cat. Spoiler: your cat shares about 96% of its DNA with a tiger, so expect a high score.

☐ +1 **It sleeps most of the day.**
Adult cats sleep roughly 12 to 16 hours a day.

☐ +1 **It purrs at you when it wants food.**
That purr hides a baby-cry-like sound (220 to 520 Hz) that owners find more urgent (McComb 2009).

☐ +1 **It ignores your birthday cake.**
Cats cannot taste sweet: the sweet-receptor gene has a chunk missing (Li et al. 2005).

☐ +1 **It knows its name and does not answer.**
Cats tell their own name from other words, but react only a little, with ears or head (Saito 2019).

☐ +1 **It hunts anything that moves.**
Free-ranging US cats are estimated to kill 1.4 to 3.7 billion birds a year (Loss et al. 2013), mostly unowned ones.

☐ +1 **It talks to you, not to other cats.**
Scientists think adult cats save most of their meowing for humans.

☐ +1 **Its tongue feels like sandpaper.**
Tiny hollow spines on the tongue wick saliva into the fur. Tiger and lion tongues were tested too (Noel and Hu 2018).

☐ +1 **It has lived with humans for ages.**
A human and a cat were buried together in Cyprus about 9,500 years ago.

YOUR VERDICT

0-2	3-4	5-6	7-8
Suspiciously normal	Tiger-curious	Tiger in a cat suit	96% tiger. The other 4% sits in the box



Sources McComb et al., Current Biology 2009 · Li et al., PLoS Genetics 2005 · Saito et al., Scientific Reports 2019 · Loss, Will, Marra 2013 · Noel and Hu, PNAS 2018 · National Geographic (Cyprus cat burial)

The tiger-cat fact sheet

What would really happen if your cat grew to tiger size. Spoiler: physics says no.



THE SCALE-UP MATHS

MATHS · MY PHYSICS REASONING

A house cat is about **4.5 kg** and a big tiger about **200 to 220 kg**. That is roughly **49 times** the mass, but only about **3.7 times** the length (length grows with the cube root of mass). A 46 cm cat body becomes about 1.7 m. Its food needs rise by only about 18 times, not 49. And muscle strength scales with area while weight scales with volume, so a giant cat would not jump 49 times higher: it would jump about the same height, and falls would hurt far more. The legendary counter leap is gone.



4.5 kg about 200 kg

96% tiger (about)

The tiger genome is 95.6% similar to the domestic cat, after splitting about 10.8 million years ago (Cho et al. 2013). Human and gorilla: 94.8%. It is DNA similarity, not "genes".

The purr twist

Tigers cannot purr the continuous way house cats do (the anatomy explanation is debated). A scaled-up house cat would keep its purr, and its baby-cry trick, at tiger volume.

Tigers miss a lot too

Tigers kill roughly once every 6.5 to 8.5 days, and hunting success is low: estimates run from about 1 in 12 to 10 to 15%. Your cat stalking the sofa is the same maths.

Cat falls

In a Berlin study of 1,125 high-rise cases, 87% of cats survived (average fall 8 to 15 m). In Greece, 93% of 207 survived falls from floors 2 to 8. A tiger-sized cat would not get that physics.

MYTH BUSTED

"Tiger pee smells like buttered popcorn." It is the binturong (bearcat) that smells like popcorn, from a compound called 2-acetyl-1-pyrroline (Duke, 2016). Not tigers.

BONUS

It never answers to its name, at any size. Scientists proved cats recognise it. It just chooses a very high altitude from which to ignore you.



The final score

Ninety-six percent tiger. The other four percent is the part that sits in the box. If your cat ticked all eight boxes, congratulations: you own a perfectly normal cat, and it has been outranking you for about 9,500 years.

Sources Cho et al., Nature Communications 2013 (tiger genome) · Duke Today 2016 (binturong) · WCS tiger kill rates; Fu Berlin and Univ. of Thessaly (falling cats) · Haldane, On Being the Right Size (1926) · National Geographic (tigers and purring)